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Comparative Analysis of Acted-Out and Real Patient Gait Parameters for Sensing Applications

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***Abstract*—Neurological conditions greatly affect patients' mobility and can considerably shorten their lifespan. However, the shortage of neurologists can limit access to essential evaluations, particularly in underserved areas. On the other hand, alternative sensing methods (e.g., videos, WiFi, mmWave), based on machine learning, can provide cost-effective solutions, and have thus gained tractions in recent years. However, these systems need large/diverse datasets for development/evaluation, collection of which can be considerably challenging from real patients. This has then made recruiting actors to replicate gait disorders a common practice, raising questions about the similarity of their gait patterns to those of real patients and the generalizability of the subsequent findings. In this paper, we methodically investigate the extent to which actors can accurately replicate gait disorders, and the corresponding impact on gait disorder classification systems, marking the first study of its kind to the best of our knowledge. Specifically, we collect videos of real patients, largely based on collaboration with a neurology center, and further recruit actors to closely act them out. By statistically analyzing key gait parameters across the two datasets, we then reveal interesting underlying disparity patterns. We further develop gait disorder classifiers to illustrate the impact of this disparity on the classification performance. Our results show interesting trends, e.g., mild gait conditions are classified more accurately than what would have been with real patients, while moderate cases are classified less accurately. Overall, the paper provides a platform for interpreting the results when recruiting actors, thus paving the way towards equitable healthcare.**

***Index Terms*—Gait disorders, WiFi sensing, Vision-based gait analysis, Smart health, Neurological disorders.**

I. Introduction

Nervous system disorders have a profound global impact, leading to approximately 11.1 million deaths and accounting for 275 million years of life lost worldwide in 2021 [1]. For instance, neurological conditions, such as dementia, Parkinson's disease, diabetic neuropathy, stroke, and multiple sclerosis, alone affected over 370 million individuals globally in 2021 [1]. Despite the severe impact of these conditions on the quality of life, there is a significant shortage of neurologists in many regions worldwide, resulting in some patients receiving medical care only after their condition has progressed significantly.

Leveraging sensing systems that use machine learning algorithms has a great potential to address the shortage of neurologists, by augmenting their assessments and providing primary evaluations to patients. This has resulted in much excitement in the research community in recent years, with a great body of work proposing to use sensors such as cameras, RF signals (e.g., WiFi, mmWave), or on-body devices for assessing gait health and detecting gait abnormalities.

For instance, [2]–[5] use joint extraction algorithms (e.g., MediaPipe Pose [6]) to classify Parkinson's severity or to detect abnormal gait from video of a person's walk. Other researchers [7], [8] expanded on this by classifying gait conditions into multiple categories. Alternatively, a Kinect sensor, which additionally captures depth information by a time-of-flight infrared sensor, was used in [9]. Similarly, researchers have employed radio frequency (RF) signals for gait monitoring and classification. For example, [10] uses WiFi to detect gait disorders including conditions such as Parkinson's, neuropathy, and dementia, and evaluates its performance against both a video-based system and assessment by neurologists. An ultra-wideband (UWB) radar at 4.3GHz was used in [11] to detect gait spasticity based on knee angles during a walk. While Wi-Flex uses WiFi to detect abnormal, non-intentional and jerky body motions such as reflexes [12].

Unlike the aforementioned studies that relied on real patient data, several work have trained and evaluated their gait disorder assessment system by recruiting healthy individuals to act out gait disorders [13]–[21]. This is mainly because collecting a large dataset from real patients can be challenging. For instance, [13], [14] developed vision-based classifiers to identify various gait disorders using acted-out data. Additionally, [15] and [16] fine-tuned a pre-trained, large image classifier using vision-based acted-out gait data. As a potential tool for monitoring rehabilitation progress in post-stroke patients, [17] trained and evaluated an attention-based architecture to classify gait into normal and three severity levels of hemiplegic gait, using video data acted out by 14 healthy participants. A 24 GHz UWB radar was used in [18] to classify five gait patterns performed by four actors. Similarly, [19] recruited 96 actors to replicate the gait of real patients from their video to train and test a classifier based on WiFi signals. [20] proposed a method for performing two neurological tests using RF signals to detect paraparesis, evaluated with simulated patient data generated by five actors. Finally, [21] used WiFi signals to detect nocturnal seizures, recruiting 20 actors to simulate tonic-clonic episodes.

Overall, the aforementioned studies have taken important steps toward an equitable and smart healthcare system. However, they employ actors to replicate gait disorders for system design and evaluation purposes. This is mainly because

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collecting large, diverse datasets from real patients is considerably challenging, as it not only requires collaboration with neurologists but also depends on patient consent to participate. Additionally, data collection from patients with gait disorders can be further complicated by their physical limitations. These issues can then prolong such studies beyond the tolerable timeframe for publication in academic circles.

To overcome these challenges, recruiting actors to simulate patient gait patterns has become common in both video-based and RF-based gait assessment studies [13]–[21]. While this has provided a more feasible path forward in this area, it does raise questions on the generalizability of the findings. For instance, to what extent would the results align with those from a system that was tested on real patients? Are there underlying trends when comparing a system developed and tested with actors to one with real patients? Would testing on real patients yield more optimistic or pessimistic outcomes? Hence, a comprehensive study is needed to explicitly assess how accurately healthy subjects can simulate gait disorders, providing a solid foundation for properly interpreting the results or adjusting the conclusions. However, no prior research has addressed this, which is the main motivation for this paper. We next explicitly discuss the contributions of the paper.

Statement of Contributions: In this paper, we investigate the extent to which actors can accurately replicate the gait patterns of patients suffering from a variety of gait disorders, and the corresponding impacts on the performance of gait disorder classification systems. More specifically, we collected video data of 30 real patients, largely based on collaboration with a neurology center. Our dataset corresponds to a range of gait-affecting conditions, including Parkinson's disease, dementia, neuropathy, post-stroke, and arthritis, with different gait disorder severity levels (i.e., mild, moderate, and severe). Similarly, we recruited actors to closely act out these gait conditions by watching the collected videos of the patients. In addition, we collect the gait of 30 healthy subjects for reference. We then extract key gait parameters from both real and acted-out datasets. By comparing these parameters across the two datasets via statistical analysis, we then reveal interesting underlying patterns. For instance, we unveil how acting results in a higher variance in the corresponding gait parameters, across all the gait parameters as well as all the gait severity levels. This increase in variance can be mainly attributed to the fact that healthy subjects cannot exactly match the severity level of a condition based on watching the video of a real patient. For instance, moderate conditions may sometimes be exaggerated, while at other times underplayed, when acted out.

Finally, we develop a number of binary classifiers (healthy vs. unhealthy gait) to illustrate the impact of disparity of gait parameters on a classification system, using a separate online dataset for training. We then evaluate their performance on both real patients and actors to explicitly show the impact of disparity of the gait parameters on the overall classification performance. Our results show interesting patterns. For instance, we shall see that mild gait conditions are classified as gait disorders with a much higher accuracy when using acted-out data as compared to real patients. However, moderate conditions are misclassified with a much higher probability as compared to the real data. This can then result in misleading conclusions when dealing with acted-out data. More specifically, testing on acted-out data can create a misleading impression that a given sensing system is capable of detecting mild gait disorders with a satisfactory accuracy. However, when the same system is tested on real patients, the accuracy would become much lower. Similarly, testing with actors for moderate conditions will give worse results than what would have been the true performance if tested on real patients.



Fig. 1. (left) Experimental setup for data collection from patients and healthy participants in a neurology clinic. (right) Experimental setup for data collection from actors.

To the best of our knowledge, this study is the first to methodically assess the disparity between the gait parameters of actors and real patients through statistical analysis, and further show its impact on gait disorder classification. It then provides a platform for interpreting the results when using acted-out data and can thus pave the way towards a smart and equitable health system for neurological gait disorders.

II. Data Collection

In this section, we introduce the three main datasets used in our analysis and discuss their collection process.

A. Data Collection from Real Patients

To collect data from real patients, we collaborated with a neurology clinic. The clinic provided us with a covered area for the experimental setup. We then marked the start and end points of a 3.9-meter straight line, as illustrated in Fig. 1 (left). The participants were patients from the neurology clinic with gait-affecting conditions, who had been diagnosed by a neurologist and agreed to participate in the experiment. Each participant was then instructed to walk back and forth twice along the line in front of the camera. This procedure allowed us to capture the body from both sides of the sagittal plane. The camera was positioned 1.9m (std=0.1m) away from the walking line at a height of 0.9m. We utilized the ultra-wide mode with an f/2.4 aperture, a 120-degree field of view, and a frame rate of 30fps.

Our partnering neurologist further categorized the severity level of the collected data into three levels: mild, moderate, and severe, based on visual assessment. We note that we did not include very mild cases (as categorized by our partnering neurologist) in our dataset since their gait closely resembles a healthy walk. Furthermore, we did not include patients that required assistive devices, such as a cane. Overall, we collected the data of 8 patients with mild gait disorders, 10 with

moderate, and 6 with severe, with the following breakdown of conditions: Parkinson's (7 patients), dementia (7 patients), neuropathy (2 patients), post-stroke (5 patients), and arthritis (3 patients). Each patient completed two back-and-forth walks.

To balance the severity levels in our real patient dataset, we augmented the collected videos with 2 mild and 4 severe arthritis patients from an online dataset of real patients [22]. This augmentation resulted in a balanced dataset of 10 patients with mild, 10 with moderate, and 10 with severe gait disorders.

Remark 1: We note that choosing a vision system as our sensing mechanism not only allows us to accurately capture the gait parameters, but also serves as a reference for actors, ensuring that they closely mimic the specific patient condition for a meaningful comparison, as we see later in the paper.

B. Data Collection from Healthy Participants

To establish a control dataset, referred to as the healthy subjects dataset, we collected data from 30 participants in the clinic, following the same experimental procedure described in Sec. II-A. These participants were individuals without any known gait disorders.

C. Data Collection from Actors

We have recruited 20 actors and tasked them with closely mimicking the gait of the patients of the dataset of part II-A. The experimental setup, configured in a conference room, is illustrated in Fig. 1 (right), where the start and end of a 3.9-meter path are marked. The camera settings matched those described in Sec. II-A, with the camera positioned 2m from the path. For each actor, we randomly selected 6 videos from our real patient dataset, 2 from each severity level (mild, moderate, and severe). Actors were shown a video of one back-and-forth walk and were informed that the patient had a gait-affecting diagnosis. They were instructed to watch each video as many times as needed to confidently act out the observed gait. The actors then performed their walk immediately after watching each video. We note that the six selected videos for each actor were presented in a random order, in order to eliminate any potential bias towards severity levels. The process relied entirely on the judgment of the actors and their perception of the video, and resulted in our acted-out dataset of 120 back-and-forth walk videos, with 40 for each severity level.

D. Human Subject Recruitment Process and Ethical Protocol

We note that our Institutional Review Board (IRB) committee has reviewed and approved this research. During the experiments, the participants were handed out a consent form that detailed the purpose of the study, experimental procedures, data collection process, and the compensation. Overall, our procedure ensured that all participants were fully informed and that the study adhered to our IRB guidelines.

III. How close Can Actors Mimic Gait Disorders - A Statistical Analysis

A thorough comparison of the walking patterns of real patients and actors requires incorporating and evaluating key gait parameters. For this purpose, we select four spatiotemporal parameters widely utilized in medical practices and/or the research community. These parameters are average gait velocity, stride length, swing time, and stance time [4], [10], [23], [24]. We consider the gait velocity as the speed of the torso. Stride length refers to the distance a foot travels from the moment it lifts off the ground to when it lands again. Swing time is the duration of this movement, while stance time is the period a foot stays on the ground during each gait cycle. In addition to these four core parameters, we calculate the first and third quartiles of gait velocity to capture broader aspects of gait dynamics, resulting in a total of six gait parameters. Other standard spatiotemporal features, such as gait cycle, step time, step length, and cadence, are highly correlated with the selected parameters. Thus, the chosen parameters provide a comprehensive representation of gait characteristics.

In the next section, we will discuss the aforementioned six gait parameters in detail, focusing on their extraction method.

A. Gait Parameter Extraction and Processing

As discussed in Sec. II, participants walk back and forth along a straight line in front of the camera. For each round-trip, we crop the videos to capture the forward and backward segments of each participant's walk separately.

We then apply MediaPipe Pose [6] to each video segment. MediaPipe Pose is a machine-learning-based pipeline for real-time joint extraction on edge devices. Using this tool, we extract the normalized image coordinates of key body joints, such as the heels, shoulders, and knees. These joints are shown in Fig. 1. The normalized image coordinates are then converted into global coordinates based on the estimated distance covered by the camera along the walking line across all horizontal pixels.

From all the extracted joints in Fig. 1, we focus on the horizontal positions of the shoulders and heels to derive the gait parameters. Therefore, throughout the rest of the paper, position refers to the horizontal position, which is aligned with the walking path discussed in Sec. II. To mitigate fluctuations caused by noise, we first apply a smoothing filter with a 0.2s window size to the shoulder and heel positions. MediaPipe occasionally misidentifies the right foot as the left, and vice versa, which introduces erroneous peaks and troughs in the position of the heel. To address this issue, we identify local extrema (peaks or troughs) that persist for less than 0.3 seconds on both sides of each extremum point and replace the position values by linearly interpolating their surrounding points.

Once the shoulder and heel positions have been denoised, we extract gait parameters from them. Each heel undergoes two phases in a gait cycle: the swing and stance phases. During the stance phase, the heel position remains stable, whereas it changes during the swing phase. The foot lift-off point is identified when the heel transitions from a stationary to a moving state, while the opposite transition marks a touch-down point. Therefore, to determine these points accurately, we calculate the variance of the heel position over a 0.4-second window before (v_b) and after (v_a) each point. Lift-off points are then detected using peaks in the function $\alpha v_a - \beta v_b$, while peaks in $\alpha v_b - \beta v_a$ indicate touch-down points (at least 0.3

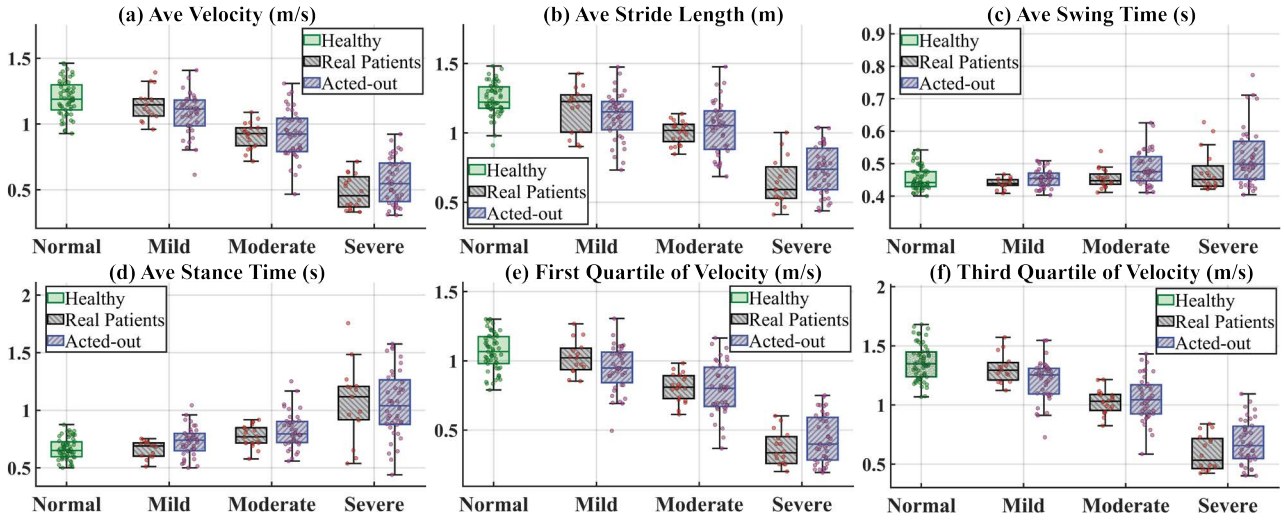


Fig. 2. Comparison of the six gait parameters for healthy subjects, real patients, and actors. See the color pdf for proper viewing.

seconds away from each other and higher than a threshold γ). The values for α, β, and γ — 4×10^3, 1.6×10^4, and 1.2 — were empirically determined from multiple walk samples.

With the shoulder and heel positions, as well as the lift-off and touch-down points, we extract six gait parameters. To remove unwanted movements at the start and end of each walk, as well as potential inconsistencies in cropping the videos, we derive the following parameters from the middle 80% of the duration of each walk.

Average Gait Velocity (m/s): Calculated by averaging the shoulder velocity, the most consistently visible joint to the camera (from the side view) connected to the torso. Average Stance Time (s): Calculated by averaging the time between heel touch-down and lift-off points. Average Swing Time (s): Calculated by averaging the time between heel lift-off and touch-down. Average Stride Length (m): Calculated as the average distance a heel travels between lift-off and touch-down. First (third) Quartile of Gait Velocity (m/s): Obtained by finding the 25th (75th) percentile of the shoulder velocities, the most consistently visible joint of the torso.

The gait parameters from the left and right sides are then averaged to represent each participant's gait characteristics. If noise prevents the detection of parameters on one side, the data available from the other side are used without averaging. To further remove very noisy samples, we exclude any back-and-forth walk when at least one of its feature values fall below the first quartile or exceed the third quartile by more than three times the interquartile range. This resulted in the removal of one round-trip walk from the healthy subject dataset, three from the real patient dataset (two from one subject and one from another), and one from the acted-out dataset.

B. Statistical Analysis and Results

We next study the key gait parameters of both real patients and actors and reveal interesting disparity patterns.

Fig. 2 illustrates the six gait parameters described in Sec. III for healthy subjects, real patients, and actors across various severity levels. The plots indicate that both the real patient and acted-out samples demonstrate a decrease in gait velocity and stride length as the severity of the gait disorder increases, while stance time and swing time tend to increase.

A notable trend observed in the gait parameters of the acted-out dataset, compared to those of real patients, is a higher variance. This increased variance can be seen across all the parameters and all the severity levels. This can be attributed to the fact that healthy actors can have a hard time exactly matching the severity level of an observed gait disorder. For instance, an actor trying to mimic a video of a patient with a moderate gait disorder may end up performing a more or less exaggerated gait disorder, resulting in a milder or a more severe acted-out case. We next further validate this hypothesis by performing Principle Component Analysis (PCA).

More specifically, we use the first two principal components of each normalized gait parameter as its representation, in order to visualize the collected data in 2D. Fig. 3(a) and Fig. 3(b) show the corresponding components for the patients and actors, respectively. We have also included the first two principal components of the healthy subjects as a benchmark.

By comparing Fig. 3(a) and (b), we can see an interesting trend. For the real patients, we can see a more clear separation along the first principal component between most severity levels, especially as the severity level gets worse. In contrast, the acted-out gait samples in Fig. 3(b) show less distinct boundaries between severity levels along the first principal component. Severe cases overlap more with moderate ones, while moderate samples mix with the mild. Additionally, some mild cases even shift towards the patients with severe gait disorders. Overall, the acted-out gait parameters demonstrate less distinct separation between severity levels compared to those of the real patients. This result is consistent with the observation made in Fig. 2.

We next investigate the difference between the first principal component of the acted-out patients and that of the real ones, comparing it to the difference between two separate walk samples of the patients. This analysis will provide a reference

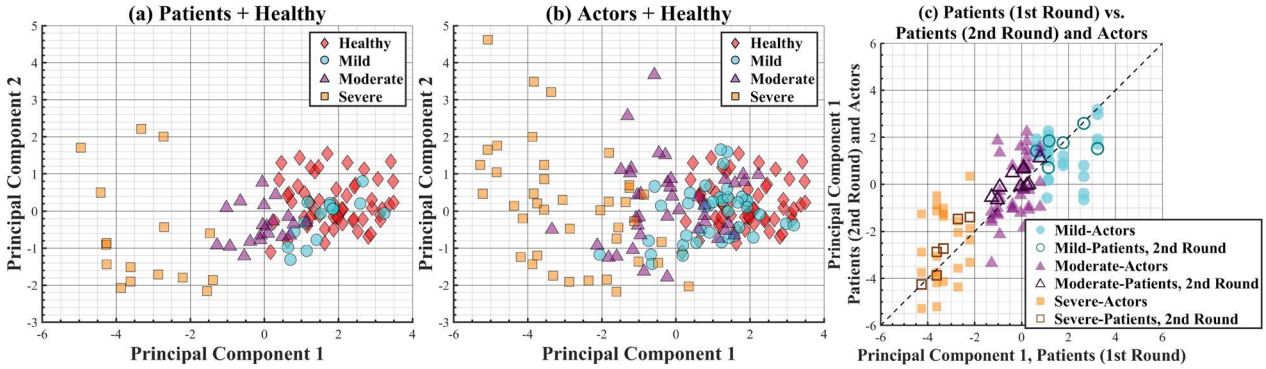


Fig. 3. (a) The first two principal components of normalized gait parameters for real patients and healthy subjects. (b) The first two principal components of normalized gait parameters for acted-out gait samples and healthy subjects. (c) Comparison of the first principal components of the normalized gait parameters between the first round (x-axis) and both the second round and acted-out data (y-axis). See the colored version for the best visualization.

to better quantify the disparity between the acted-out and real patient's data. Fig. 3(c) illustrates the first principal component (PC1) of the normalized gait parameters. More specifically, the figure compares the PC1 of each acted-out sample (y-axis) to that of the first round-trip walk from the corresponding patient (x-axis). For baseline comparison, we then plot the PC1 of the second round-trip walk from each patient (y-axis) against that of their first round (x-axis). We can clearly see that the first round walk has more similarity to the second round walk than to the acted-out data. To better quantify this, we then calculate the Pearson correlation coefficient for the PC1 data of the two respective groups. The correlation between the PC1 of the acted-out data and that of the first round-trip walk for the real patients is 0.77, whereas the correlation between the PC1 of the second and first round-walks is 0.96. These results highlight the greater disparity in the acted-out gaits as compared to the two different rounds of real patients' walks.

IV. Impact of Acting on Classification Performance

In the previous section, we showed the impact of acting gait disorders on the gait parameters. We next discuss how this translates to the performance of a classifier that is tested on acted-out data.

A. Generating a Training Dataset

In order to train a classification pipeline and test the impact of acting and the aforementioned observed disparity in the gait parameters of actors vs. real patients, we have used an online dataset of real patients. More specifically, [22] includes side-view videos of individuals with Parkinson's disease, arthritis, and healthy subjects. In this dataset, participants walked along a 4.5m straight line back and forth once, and were recorded by a camera positioned 8m away from the line. Our training dataset includes 20 healthy subjects, 13 Parkinson patients (6 mild and 7 moderate), and 44 arthritis patients (13 early, 20 moderate, and 11 severe).

B. Training Gait Disorder Classifiers

In this section, we evaluate the classification accuracy of SVM, KNN, and MLP classifiers in identifying subjects with a gait disorder on both real patient and acted-out datasets. The models were trained on the online training dataset described in Sec. IV-A, which includes 87 subjects (30 healthy and 53 with gait disorders), using the six features outlined in Sec. III. The goal of the classifiers is to provide a binary label indicating healthy vs. unhealthy gaits.

More specifically, we train an SVM model with a linear kernel and a KNN classifier with five neighbors using the normalized features of the online dataset. Additionally, a shallow MLP with a single hidden layer consisting of 6 perceptrons and a ReLU activation function was trained on the first three principal components of the normalized features. The model was learned using ADAM, with a learning rate of 0.001, a batch size of 8, and trained over 140 epochs, in PyTorch.

We then test the trained classifiers on both the real patient dataset as well as the acted-out ones. It is noteworthy that the few online data samples that were added to augment the patient and acted-out data as part of the analysis of the previous section (as discussed in Sec. II) are removed from the testing sets in order to ensure no bias.

C. Comparison of Classification Performances

Table I shows the breakdown of the results for both real patient and acted-out datasets as a function of different severity levels. A few interesting trends can be observed. All the classifiers have a considerably lower accuracy for moderate cases in the acted-out gait samples, as compared to real patients. However, the classification accuracy for mild cases is considerably higher for the acted-out cases as compared to the real patients. In other words, testing with acted-out patients can result in an over-optimistic performance for mild cases while the performance for moderate cases can become over-pessimistic. This behavior can be justified by the analysis of Fig. 2 and 3. Specifically, the higher variance of the gait parameters for the acted-out dataset, observed in Fig. 2, resulted in the mixing of the data points of different severity levels in the principal component analysis of Fig. 3. In other words, some of the mild cases were acted out more like moderate cases due to the difficulty of exactly matching the observed severity level. Given that moderate cases are easier to differentiate from healthy cases, this then results in mild acted-out samples being classified as gait disorder with a higher probability than what it would have been with real patients.

TABLE I
CLASSIFICATION ACCURACY OF DIFFERENT BINARY CLASSIFIERS WHEN TESTED WITH HEALTHY SUBJECTS, REAL PATIENTS, AND ACTORS, AND AS A FUNCTION OF DIFFERENT SEVERITY LEVELS.

Classifier	Healthy	Unhealthy		Gait Disorder Severity Level					
				Mild		Moderate		Severe	
		Real Patients	Acted-Out	Real Patients	Acted-Out	Real Patients	Acted-Out	Real Patients	Acted-Out
SVM	75%	80%	76%	31%	52%	100%	80%	100%	100%
KNN	73%	82%	75%	38%	52%	100%	78%	100%	100%
MLP	83%	76%	72%	24%	44%	95%	77%	100%	100%

Similarly, some moderate cases are acted out more like mild cases, while others resemble more severe cases, as evidenced by Fig. 2. However, given that mild cases are hard to identify as gait disorders, this then results in an over-pessimistic performance when testing the classifiers with moderate acted-out cases. Finally, severe cases result in a comparable classification accuracy for both acted-out and real patients, as expected. It is worth noting that the overall classification accuracy over all the severity levels remains similar when comparing the acted-out and real patients as it averages over different severity levels. For instance, the overall classification accuracy, averaged over all the severity levels, is 83% for healthy subjects, 76% for real patients, and 72% for acted-out samples for the MLP (averaged over 20 training iterations). However, this can be misleading if one does not consider the detailed breakdown performance over different severity levels.

Remark 2: In this section, the classifiers were trained on real data while tested on acted-out as well as real patients' data. We note that we have observed similar disparity trends if the classifiers were instead trained on acted-out data, while being tested on both actors and real patients.

Overall, the results indicate that the classification performance on acted-out samples can be vastly different from that on the real patients. The disparity is highly correlated with the severity level of the gait condition, as we have shown. As such, any results obtained with actors should be interpreted differently, depending on the severity level of the gait disorder.

V. CONCLUSION

This paper considered neurological gait disorders and methodically analyzed the disparity between the gait parameters of real patients and those simulated by actors. More specifically, we collected the videos of real patients with five different gait disorders in a neurology clinic, and further recruited healthy subjects to replicate them. Our statistical analysis of the two datasets then revealed interesting underlying patterns such as a higher variance in the gait parameters for the acted-out dataset. This disparity affected the performance of the subsequent gait disorder classifiers, showing a higher classification accuracy for mild acted-out cases and lower for moderate ones, as compared to the accuracy when tested with real patients.

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